Jiachi ZHANG

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EDUCATION

South China University of Technology 09/2023 - 06/2027

Qualification: Bachelor of Engineering

Major: Information Engineering

Average Score: 84.52/100

Core Courses: *Signals & Systems, Analog Electronics, Digital Electronics, Digital Signal Processing, Principle of Communications, Microcomputer System and Interface Technology, Programming in C++*

SUMMER STUDY PROGRAM

École Polytechnique de l'Université de Nantes, France 07/2025

Core Courses: *Français, théorie de l'information*

INTERNSHIP EXPERIENCE

Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences 10/2025 - Present

Position: *Visiting Research Intern*

- Carried out independent research by identifying research gaps, developing research hypotheses, designing experimental protocols, and conducting hands-on experiments and data analysis.
- Compiled academic materials by reviewing relevant literature and drafting concise and structured experimental summaries.
- Authored and revised the thesis, ensuring a clear, logically rigorous scientific narrative grounded in experimental results.

RESEARCH EXPERIENCE

Gain-Aware Prediction-Space Recursive Controller for Lightweight Polyp Segmentation

Submitted to ICONIP 2026 | **Jiachi Zhang** et al.

• Formulated segmentation refinement as prediction-space recursive correction with only 189 parameters and negligible GMACs overhead.

• Designed a three-stage recursive controller: Evidence Encoding, Gain-Aware State Update, State-Guided Residual Correction.

• Evaluated across 7 lightweight backbones on 4 datasets, consistently outperforming training-side baselines (+LL/+DS/+Mutation/+LoMix) and +Harmonizing.

RIGS-Refiner: Risk-Guided Recursive Refinement in Prediction Space for Polyp Segmentation

Submitted to APSIPA ASC 2026 | **Jiachi Zhang** et al.

• Proposed a lightweight post-refinement plugin with 519 parameters combining image priors, prediction cues, risk-guided update, and residual correction.

• Tested on PraNet and SegFormer-B0 across 4 datasets, achieving favorable efficiency–accuracy trade-offs against SegFix, rNCA, and CascadePSP (×2600 parameter reduction).

Glomerular Pathological Image Detection via YOLO11 with Stain Normalization 04/2024–03/2025

• Designed color clustering and transfer modules to mitigate staining variance; built YOLO11-based detection framework with EMA attention and Focal Loss, achieving 94% mAP.

PROJECT EXPERIENCE

Macroeconomic Indicator Forecasting System via SAITS and Time-Series LLMs 07/2025 - 08/2025

- Adopted the SAITS algorithm to conduct multivariate time-series imputation, and reconstructed high-fidelity continuous feature spaces under limited labeled data.
- Integrated TimeLLM and ChatTS, designed cross-modal alignment modules to capture temporal evolution trends and enhance prediction accuracy.

- Constructed a modular training and evaluation system, and developed high-precision interpretable time-series models, securing top 10 in the E Fund FinTech Challenge.

FPGA-Based Shooting Confrontation Game 03/2026

- Developed a dual-mode FPGA shooting game supporting 1280×800@60fps HDMI output, infrared control and real-time score display.
- Designed HDMI display drivers and SDRAM caching logic, and implemented AABB-based collision detection for game objects.
- Used finite state machines to control seven game states and ensure synchronization between peripherals and display modules.
- Completed project validation and received the Excellent Course Design award for outstanding academic performance.

EXTRACURRICULAR EXPERIENCE

President, Art Troupe, Youth League Committee 11/2024 -11/2025

- Participated in the production and singing of the department song, and performed in the college band on various occasions.
- Managed and coordinated internal departmental affairs, and took charge of the college graduation gala as a key organizer for multiple times.

Study Representative, Class Committee 09/2023 - 10/2025

- Collected and distributed daily assignments, and completed the liaison work for academic affairs.
- Communicated actively between teachers and classmates, and released official university notifications in a timely manner.

HONORS & AWARDS

3rd Prize Scholarship, South China University of Technology 12/2025
Merit Student, South China University of Technology 12/2025
Advanced Individual in Student Work, South China University of Technology 09/2025
Active Participant in Student Work, South China University of Technology 09/2025
10th Place, Pazhou Algorithm Competition - E Fund Cup FinTech Challenge 08/2025
Outstanding League Cadre, South China University of Technology 05/2025

ADDITIONAL SKILLS

Language Ability: French (basic), English (fluent, can be used as a studying language)
Programming & Development: Python, PyTorch, Keil, MATLAB, Vivado